

Lab: Building Your Inventive Learning Plan

Objective

Assess your current inventive skill profile across perception, cognition, action, and learning. Then design a personalized deliberate practice plan, conduct a mini patent analysis exercise, and explore one biomimicry connection to your invention idea. By the end of this lab, you will have a concrete, actionable plan for systematically improving your inventive capacity.

Materials / Prerequisites

- Your own invention idea or project
- Internet access (for patent search)
- Paper and pen
- A timer
- Completed reading of Module 06: Learning to Invent

Part 1: Inventive Skill Self-Assessment (12 minutes)

Step 1 (8 min): Rate your current skill level in each inventive competency area. Be honest — the goal is to identify where deliberate practice will have the highest impact.

Skill Area	Specific Competency	Rating (1-10)	Evidence for This Rating
Perception	Noticing prediction errors / seeing what others miss		

Write your response here

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Write your response here

| | **Perception** | Cross-domain pattern recognition |

Write your response here

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| | **Perception** | Detecting latent user needs |

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Write your response here

|| **Cognition** | Analogical reasoning |

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|| **Cognition** | Generating multiple alternatives (cognitive fluency) |

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Write your response here

|| **Cognition** | Systematic problem solving (TRIZ or similar) |

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Write your response here

| | **Action** | Rapid prototyping / building quickly |

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Write your response here

| | **Action** | Sketch modeling / physical externalization |

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Write your response here

| | **Action** | Testing / running experiments |

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Write your response here

| | **Learning** | Extracting lessons from failures |

Write your response here

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Write your response here

| | **Learning** | Studying others' inventions |

Write your response here

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Write your response here

| | **Learning** | Maintaining a learning journal |

Write your response here

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Write your response here

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Step 2 (4 min): Identify your three strongest competencies and your three weakest:

Strongest: 1.

Write your response here

1.

Write your response here

1.

Write your response here

Weakest: 1.

Write your response here

1.

Write your response here

1.

Write your response here

What type of learning does each weakness require (parameter, structure, or policy)?

Write your response here

Part 2: Patent Analysis Exercise (15 minutes)

The global patent database contains over 100 million documents, each one a structured record of an inventive insight. Reading patents through the Active Inference lens transforms them from dry legal documents into rich case studies of how inventors identified the limitations of existing models and constructed better ones. This exercise gives you practice with a skill that will serve you throughout your inventive career: the ability to extract structural principles from specific inventive solutions and add them to your own cognitive toolkit.

Step 1 (5 min): Search for a patent related to your invention domain. You can use Google Patents (patents.google.com), the USPTO (patft.uspto.gov), or any other patent database. Find one patent that is relevant to the problem your invention addresses.

Patent number or title:

Write your response here

Inventor(s):

Write your response here

Year:

Write your response here

Step 2 (5 min): Read the patent through the Active Inference lens and extract the following:

The prior model (Background of the Invention) — What was the state of knowledge before this invention? What generative model did existing solutions assume?

Write your response here

The prediction error (Problem to be Solved) — What failure of the prior model motivated this invention? What did the prior model predict incorrectly?

Write your response here

The updated model (Solution) — What new generative model does this invention embody? How does it resolve the prediction error?

Write your response here

Step 3 (5 min): Extract the inventive principle underlying this patent. State it as a general rule that could apply beyond this specific domain:

The inventive principle:

Write your response here

One other domain where this principle could apply:

Write your response here

What did you learn about inventive thinking from analyzing this patent?

Write your response here

Part 3: Biomimicry Exploration (12 minutes)

Nature has been solving engineering problems for 3.8 billion years through the iterative process of natural selection — a process that is, at its core, a form of Active Inference operating at evolutionary timescales. Every organism represents a refined generative model for surviving in its environment. Biomimicry taps into this vast library of tested solutions by identifying structural analogies between biological and engineering challenges. The key is to think in functional terms: do not ask "What animal looks like my invention?" but rather "What organism faces the same functional challenge as my invention?"

Step 1 (4 min): Describe the core challenge of your invention in functional terms (what it needs to DO, not what it needs to BE):

Functional challenge:

Write your response here

Step 2 (4 min): Identify at least two biological organisms or natural systems that face a structurally similar functional challenge:

Organism / Natural System	The Similar Functional Challenge	How Nature Solves It

Write your response here

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Write your response here

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Write your response here

Write your response here

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Write your response here

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Step 3 (4 min): Select the most promising biological analogy and describe how the biological principle could be applied to your invention:

Selected organism/system:

Write your response here

The structural principle (not the surface form):

Write your response here

How this principle could be applied to my invention:

Write your response here

What would my invention look like if it worked the way this organism works?

Write your response here

Part 4: Design Your Learning Plan (15 minutes)

A learning plan for invention is fundamentally different from a learning plan for academic study. Academic learning optimizes for knowledge acquisition — absorbing and retaining information. Inventive learning optimizes for capability expansion — being able to do things you could not do before. This requires all three modes: parameter learning (getting better at what you already do), structure learning (developing new ways of understanding problems), and policy learning (acquiring new hands-on skills). The most common learning imbalance among inventors is over-emphasis on parameter learning (refining existing expertise) at the expense of structure learning (developing new frameworks) and policy learning (acquiring new practical skills).

Step 1 (5 min): Based on your self-assessment in Part 1, select one weakness from each category (perception, cognition, action) to focus on:

Category	Weakness to Address	Type of Learning Needed	Why This Matters for My Invention
Perception			

Write your response here

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Write your response here

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Write your response here

| | Cognition |

Write your response here

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Write your response here

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Write your response here

| | **Action** |

Write your response here

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Write your response here

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Write your response here

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Step 2 (5 min): Design a specific deliberate practice exercise for each weakness:

Perception practice (what you will do, how often, how you will get feedback):

Write your response here

Cognition practice (what you will do, how often, how you will get feedback):

Write your response here

Action practice (what you will do, how often, how you will get feedback):

Write your response here

Step 3 (5 min): Create a weekly learning schedule:

Day	Activity	Duration	What I Will Learn
Monday			

Write your response here

Write your response here

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Write your response here

|| Tuesday |

Write your response here

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Write your response here

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Write your response here

|| Wednesday |

Write your response here

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Write your response here

|| Thursday |

Write your response here

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Write your response here

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Write your response here

|| Friday |

Write your response here

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Write your response here

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Write your response here

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How will you track your progress? What will you use as a learning journal?

Write your response here

Part 5: The Learning Spiral Check (6 minutes)

The learning spiral — perception, cognition, action, learning — must cycle continuously for inventive skill to develop. Most inventors have a preferred phase where they spend disproportionate time: some love reading and researching (perception), others love analyzing and planning (cognition), others love building (action), and some love reflecting and writing (learning). The phase you neglect is often the one that would contribute most to your growth, precisely because it exercises cognitive capacities your preferred phase does not engage. This brief self-assessment identifies your imbalance and prescribes a corrective action.

Step 1 (3 min): Reflect on your recent inventive work. Which phase of the learning spiral (perception, cognition, action, learning) do you spend the most time in? Which do you neglect?

Phase I spend most time in:

Write your response here

Phase I neglect:

Write your response here

Why:

Write your response here

Step 2 (3 min): Describe one specific step you will take this week to activate the neglected phase:

Write your response here

Discussion / Debrief Questions

1. Which of the three learning modes (parameter, structure, policy) is most important for your current inventive work? Why? Is it the same mode you naturally practice, or do you need to deliberately shift?
2. The patent analysis exercise asks you to read patents as Active Inference artifacts. Did this framing change how you read the patent? What did you learn about the inventor's thinking process that a conventional reading would have missed?
3. Biomimicry requires translating biological principles to engineering. Did you find it easy or difficult to identify biological analogies for your invention challenge? What does this tell you about the breadth of your current generative model?
4. Deliberate practice requires targeting weaknesses, which is inherently uncomfortable. How will you motivate yourself to practice skills you are bad at rather than skills you enjoy?
5. Edison maintained daily experiment quotas. What is the equivalent for your inventive practice? How many learning cycles (perception-cognition-action-learning) per week would constitute adequate practice?